Submitted by:

mariyam Kashif

F25-BSAI-0022

09-10-2026

Assignment # 01

Submitted to:

Sir Usama

CamScanner

Assignment #01

Question #01

(a): Propositional Logic

Let:

A= Ali passes the midterm exam.

B= Ali passes the final exam.

C= Ali passes the course.

D= Ali is eligible for a certificate.

i. $(A \wedge B) \rightarrow C$

ii. $C \rightarrow D$

(b): Interferance Rules

Since A and B are given:

- $A \wedge B$

- $(A \wedge B) \rightarrow C$
- $C \rightarrow D$

Ali is eligible for the certificate.

Question #02

(a): Equivalent Form

We know:

- $P \rightarrow Q \equiv \sim P \vee Q$
- $Q \rightarrow R \equiv \sim Q \vee R$

Therefore:

$$(\sim P \vee Q) \wedge (\sim Q \vee R)$$

Using only $\sim$ and $\vee$:

$$\sim(\sim(\sim p \vee Q) \vee \sim(\sim Q \vee R))$$

(b): Logical Equivalence

Logical equivalence means that two logical expressions have the same truth value for all possible combinations of their propositions.

CamScanner

It is represented by ≡.

(c): Determine whether $P \to Q$ is logically equivalent to $\sim Q \to \sim P$.

Yes, they are logically equivalent.

Because $\sim Q \to \sim P$ is the contrapositive of $P \to Q$.

Therefore:

$P \to Q \equiv \sim Q \to \sim P$

Both expressions have the same truth values in all cases.

Question # 03

(a): $(P \vee Q) \wedge \sim P$

P	Q	P∨Q	~P	(P∨Q)∧~P
T	T	T	F	F
T	F	T	F	F
F	T	T	T	T
F	F	F	T	F

This expression is neither a tautology nor a contradiction. It is a contingency because it is true in one case and false in the others.

Question # 04

(a): identification

- **Sensors:** Dirt sensor, obstacle sensor, distance sensor, bump sensor, battery sensor.
- **Actuators:** Wheels, vaccum motor, brushes.
- **Percepts:** Dirt, obstacles, walls, distance and battery level.
- **Possible Actions:** move forward/ backward, turn left/right, clean, stop and return to charging station.

CamScanner

(b): PEAS Description

PEAS	Discription
P – Performance	Cleanliness, efficiency
E – Environment	Rooms, floor, furniture
A – Actuators	Wheels, vaccum, brushes
S – Sensors	obstacle, battery sensors, distance

if

(c): classify its environment

- Partially observable
- Stochastic
- Sequential
- Dynamic
- Continuous

Question # 05

P: It is raining.
Q: The road is wet.
R: The school is closed.

CamScanner

(a): Statements

i - $P \rightarrow Q$

ii - $R \leftrightarrow P$

iii - $\sim Q \rightarrow \sim P$

(b): Truth Table

$(P \rightarrow Q) \wedge (Q \rightarrow R)$

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$(P \rightarrow Q) \wedge (Q \quad R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

CamScanner